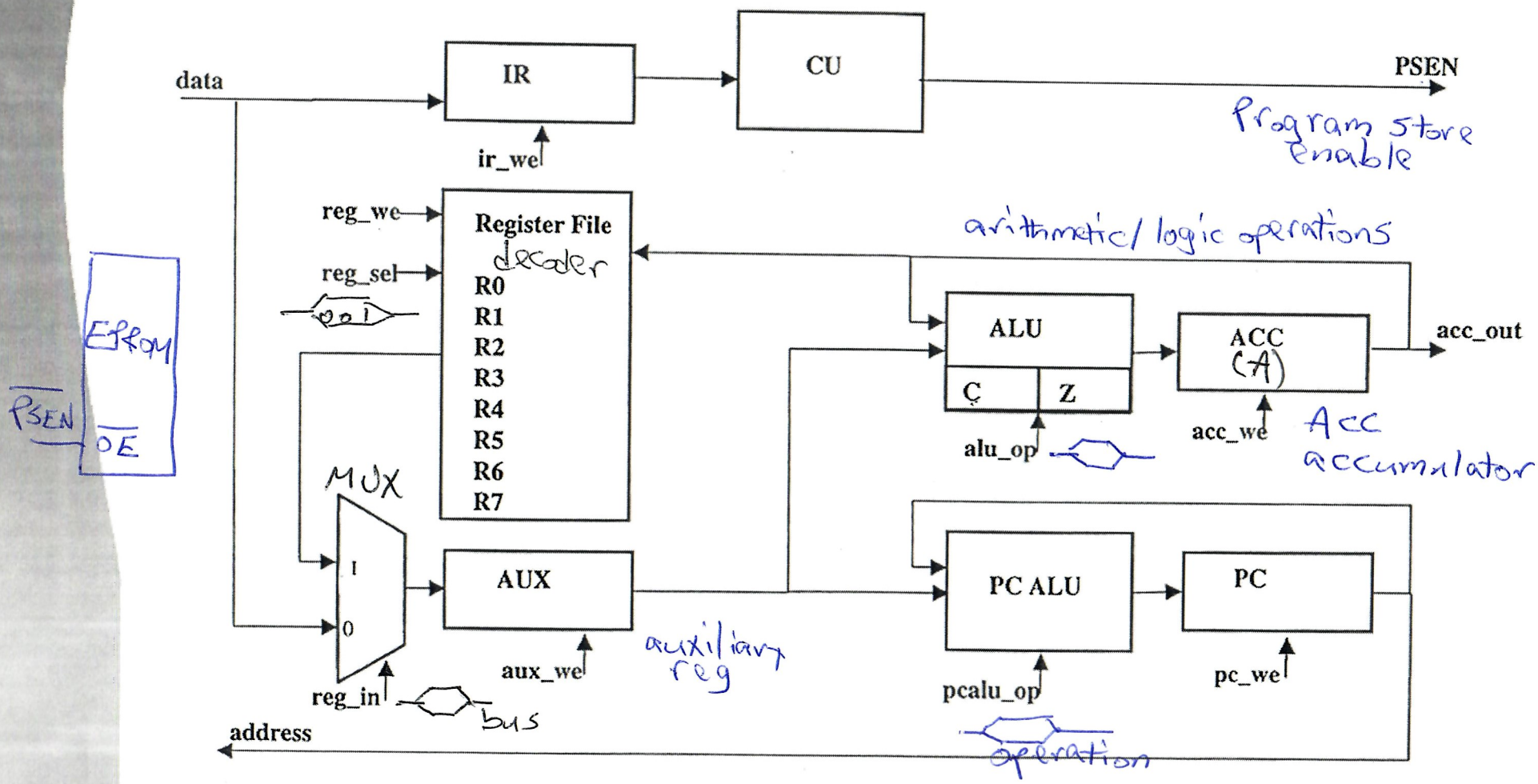
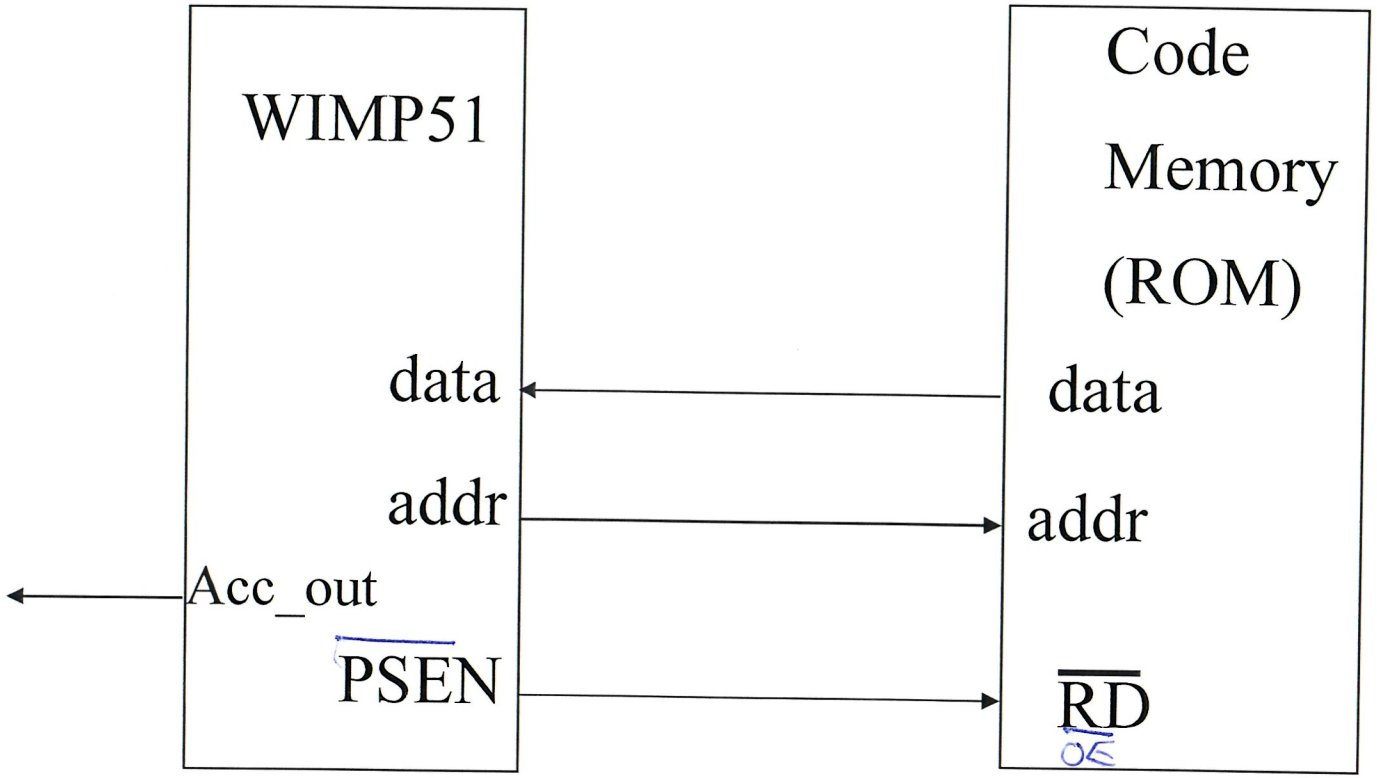
WIMP51 Architecture



8 bit microprocessor

ALU - arithmetic logic Unit

WIMP51: Outside world



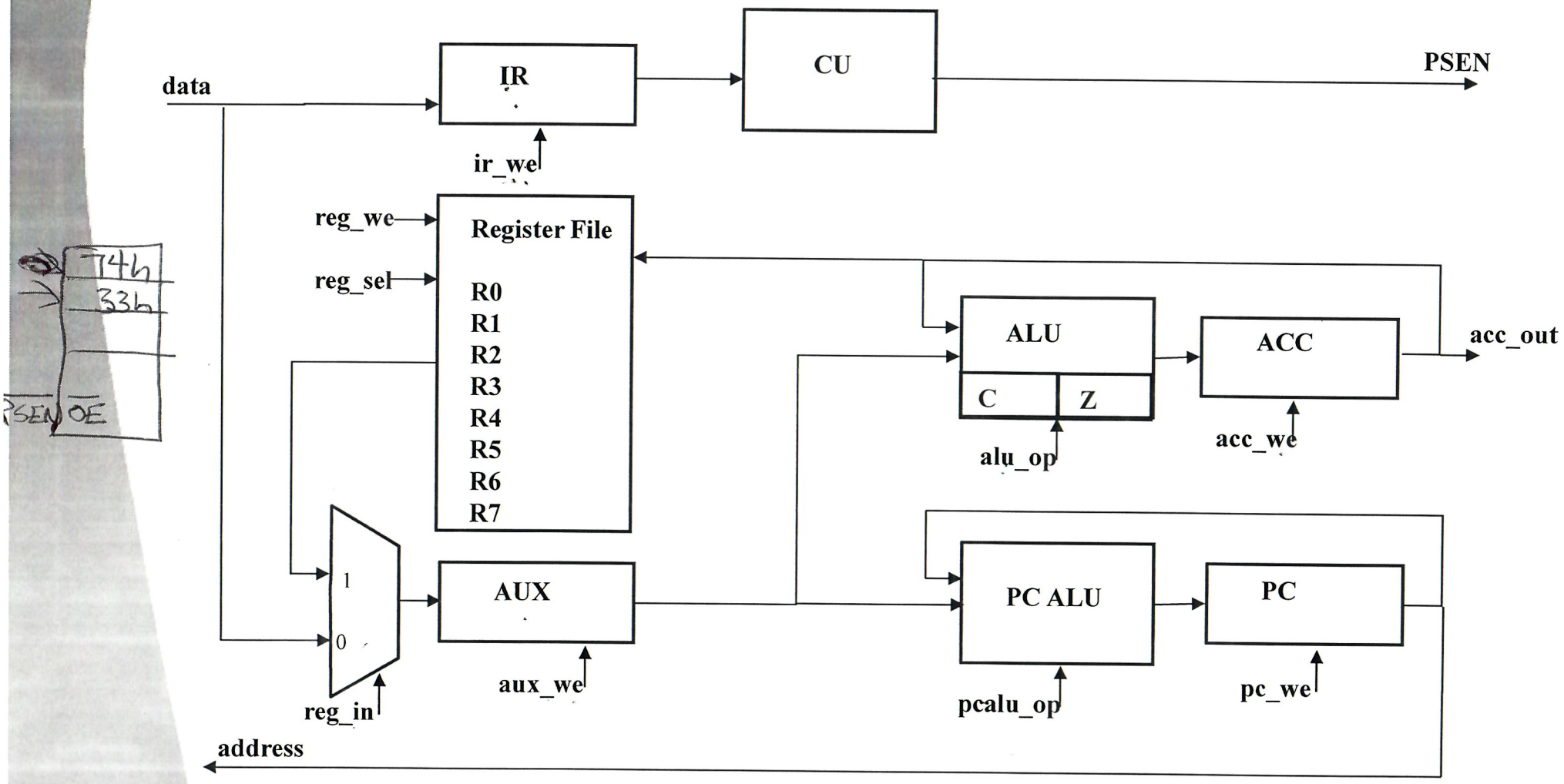
Wimp51 Instructions

MOV A, #D	01110100	dddddddd	A<=D
ADDC A, #D	00110100	dddddddd	C,A<=A+D+C
MOV Rn, A	11111nnn		Rn<=A
MOV A, Rn	11101nnn		A<=Rn
ADDC A, Rn	00111nnn		C,A<=A+Rn+C
ORL A, Rn	01001nnn		A<=A or Rn
ANL A, Rn	01011nnn		A<=A and Rn
XRL A, Rn	01101nnn		A<=A xor Rn
SWAP A	11000100		$A<=A_{(3-0)}\&A_{(7-4)}$
CLR C	11000011		C<=0
SETB C	11010011		C<=1
SJMP rel	10000000	aaaaaaaa	PC<=PC+rel+2
JZ rel	01100000	aaaaaaaa	PC<=PC+rel+2 if Z

A
Acc

JZ → Jump if Z=1

WIMP51 microprocessor



Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction MOV A, #33h Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below.

MOV A, #33h 74h 33h

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

The WIMP51 architecture schematic is available in the notes provided to you for this exam.

